

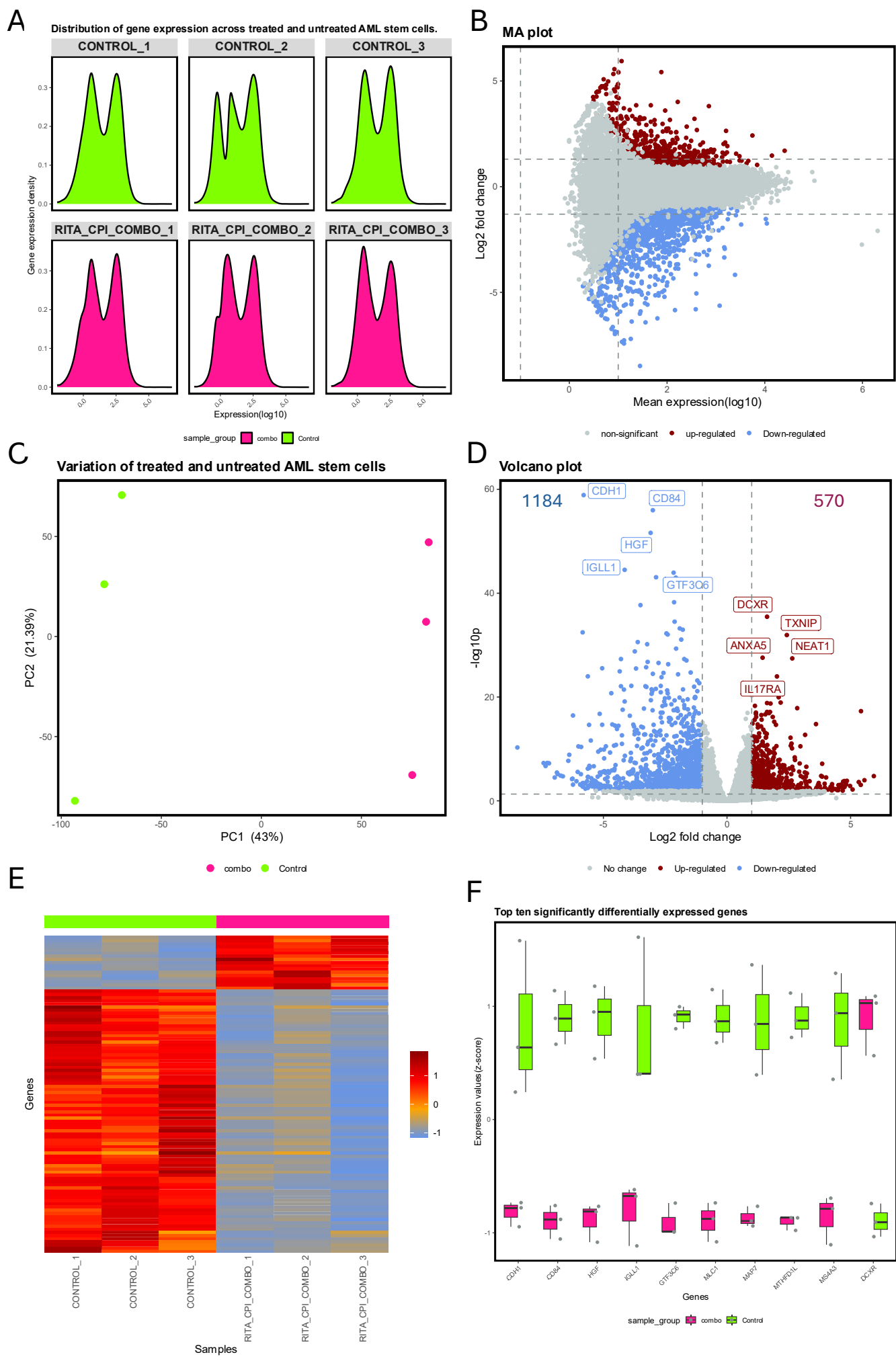


Figure 1. Quality control and visualisation of differential gene expression in untreated and treated AML stem cells (combination therapy of RITA and CPI-203).

(A) Expression density plots illustrating the distribution of gene expression in treated (pink) and untreated AML stem cells (green). The y-axis shows the density, and the x-axis shows the log10 mean expression. **(B)** MA plot displaying the mean expression levels on x-axis of downregulated (blue) and upregulated genes (maroon) shown on the Y axis (log2 fold-change), with dashed lines as cut-offs. **(C)** PCA plot showing clustering of treated (pink) and control samples (green) based on PC1 on x-axis and PC2 on y-axis with their variance percentage indicated. **(D)** Volcano plot illustrating differential gene expression between treated and untreated samples. x-axis = log2 fold change, y = -log10p with dashed lines as cut-offs for significance (p=0.05) and 1 and -1 log fold change for up or down gene regulation. The top five significantly upregulated and downregulated genes have been highlighted. **(E)** Heatmap of 100 significantly differentially expressed genes(p<0.05) across untreated and treated AML stem cell samples with a colour gradient indicating relative expression intensity . X-axis represent samples and y-axis individual genes **(F)** Boxplot overlayed with jitter plot of the top 10 significantly regulated genes, x-axis = genes, y-axis = gene expression values (z-score).

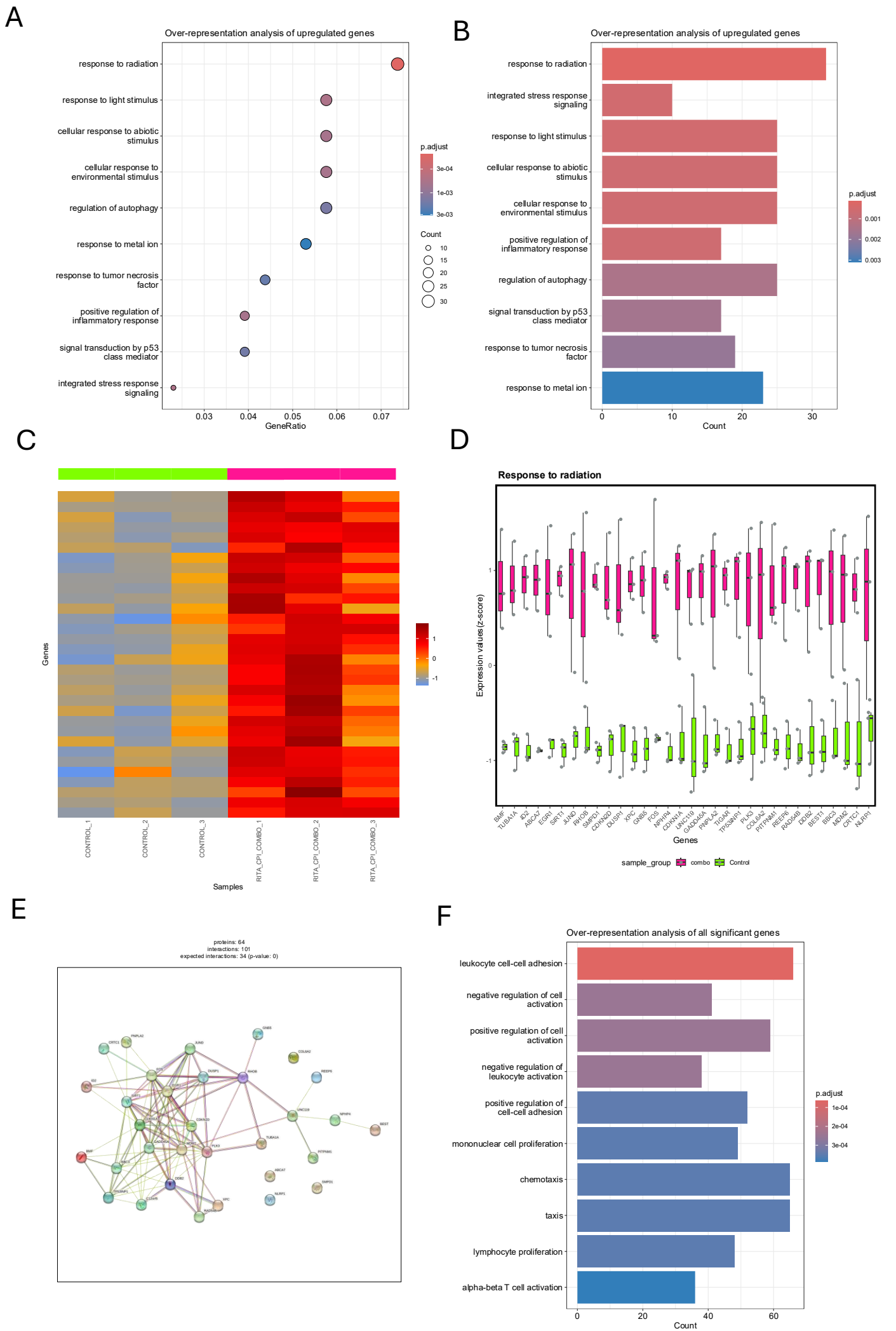


Figure 3. Biological processes: over-representation (ORA) analysis of significantly upregulated genes in the treated AML stem cells (RITA and CPI-203 treatment) and ORA of all significant genes

(A) Dot plot showing the top 10 enriched Gene Ontology (GO) terms. Each dot represents a GO term; dot size corresponds to the number of overlapping genes, and colour intensity reflects the degree of statistical significance (p.adjust). **(B)** Bar plot of the top 10 GO terms ranked by decreasing significance, with the x-axis indicating the number of overlapping genes. **(C)** Clustered heatmap illustrating expression patterns of genes within the most significantly enriched GO term, response to radiation, across treated and untreated AML samples. X-axis represent individual samples and Y axis represent genes. **(D)** Boxplot overlaid with jitter points showing expression levels of genes involved in the response to radiation GO term across conditions: treated (pink) and untreated (green). **(E)** A STRING depicting previously reported interactions among the genes in the response to radiation enriched pathway. **(F)** Bar plot showing over-representation analysis of the top 10 Gene Ontology (GO) of all significantly differentially expressed genes.